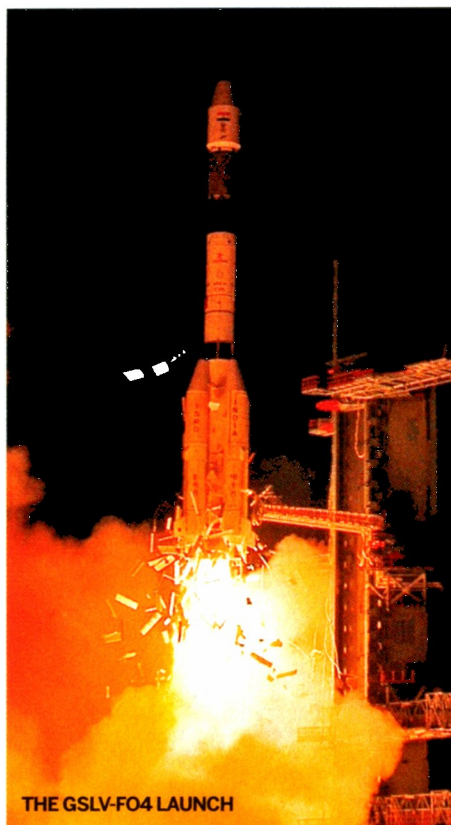


ily and burns without visible flames, the LPSC built its own leak-monitoring devices. Another powerful local cryogenic engine is being readied for the next GSLV rocket launch next year. However, the first stage of both the PSLV and the GSLV is still a solid propellant motor and the second stage is the liquid engine Vikas, with the cryogenic engine being the third stage.

Former department of science and technology secretary and head of the Vikram Sarabhai Space Centre, Thiruvananthapuram, Vasant Gowardiker, worked on cryogenic engine development in ISRO in the '70s, setting up the Cryogenic Techniques Project (CTP). It was like a seed sown to nurture the cryo programme, worth less than Rs 4 crore then. Even at that time, making liquid hydrogen was risky business—liquid oxygen could be easily sourced—but the idea was to follow up on Sarabhai's make-it-in-India dream. Some scientists blame the late Satish Dhawan and others close to him for not pursuing the Sarabhai dream. Even at that time, recalls Gnana Gandhi, the industry and private sector chipped in with their help, "so after all it was a good thing that the Americans tightened the screws on the Russians". Before it undertook development of cryogenic engines, LPSC's key mission was to develop liquid-fuelled engines.

The GSLV was meant to match the range of launchers like Ariane 4, Delta II, Long March 2 and Soyuz-U and allow India to negotiate business on a bigger scale. According to the non-profit Space Foundation, the global space business grew to \$261.6 billion in 2009, expanding 7 per cent over 2008 and 40 per cent over the past five years. The industry continued expanding in early 2010, reflect-



ISRO's ambitious project now faces a reality check

April 18, 2001 (GSLV-D1): First developmental flight of Geosynchronous Satellite Launch Vehicle (GSLV) with a Russian cryogenic engine.

May 8, 2003, GSLV-D2: Second developmental flight orbits an experimental communication satellite GSAT-2.

Sept 20, 2004, GSLV-F01: First operational flight launches Edusat satellite.

July 10, 2006, GSLV-F02: Fourth flight carrying Insat-4C fails

Sept 2, 2007, GSLV-F04: Lobs INSAT-4C replacement into orbit.

April 15, 2010, GSLV-D3: Sixth flight, with a made in India cryogenic stage fails.

ing greater demand for space-related products and services—including low-cost GPS hardware embedded in cars and phones, communications services, and control of a growing number of unmanned aerial vehicles. The variety and number of activities is likely to grow, as space products and services are integrated deeply into consumer electronics and daily necessities. Commercial satellite services increased by 8 per cent to \$90.58 billion in 2009, representing about 35 per cent of the global space economy; while global government spending on space increased 16 per cent in 2009 to \$86.17 billion, says the report. Spending on space infrastructure, including launch services, spacecraft manufacturing and

ground equipment totalled \$83.63 billion in 2009, or about 32 per cent of the overall market. Launch rates increased 42 per cent from 2005 to 2009 with India, North Korea and South Korea each accounting for less than 4 per cent.

Rocket launchers, like the latest GSLV, are also meant to boost the efforts of agencies like Antrix Corporation, ISRO's commercial arm that hopes to cross the Rs 1,000-crore mark in turnover this fiscal and is sitting on orders worth Rs 3,000 crore in the pipeline. ISRO's budget has zoomed into a higher orbit, and its current allocation at Rs 5,778 crore and sourcing funds is apparently no constraint considering the next ambitious endeavour is for a manned space flight programme which alone would cost Rs 13,000 crore. But this mission will be delayed for ISRO will have to prove its indigenous cryogenic technology even though it still has two Russian-made engines in its inventory. Given the nature and cycle of success and failure that space programmes face, ISRO will bounce back before long. ■

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K. R. SRIDHARA MURTHI

MANAGING DIRECTOR, ANTRIX CORPORATION

